

MVDR-Consistent Graph-Attention Beamforming with Local Conformer for Multichannel Speech Enhancement

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Abstract—We propose an end-to-end multichannel speech enhancement system that integrates dynamic spatial filtering, efficient temporal encoding, and complex-mask refinement. Dynamic spatial filtering is realized by a dual-transform graph-attention beamformer that produces real, normalized per-time-frequency spatial weights on a microphone graph. Under narrow-band processing, these real convex weights provide an MVDR-inspired normalized magnitude profile of the MVDR numerator, while the downstream complex-mask head handles residual phase/sign and reverberation-related errors. Efficient temporal encoding is implemented by a lightweight Conformer with localized self-attention, depthwise-separable convolutions, and low-rank projections, yielding bounded latency. A compact U-Net then predicts a complex mask applied to the spatially filtered stream, and the waveform is reconstructed via inverse short-time Fourier transform with a phase-constrained magnitude loss. Experimental results demonstrate consistent gains in speech quality and intelligibility over strong baselines while maintaining a low computational footprint.

Index Terms—Multichannel speech enhancement, graph attention beamforming, MVDR consistency, localized Conformer, complex ratio mask.

I. INTRODUCTION

Speech enhancement (SE) aims to recover target speech from noisy mixtures and remains a central challenge in signal processing. While single-channel methods lack spatial cues, multichannel approaches exploit inter-microphone differences (time delays and level disparities) to suppress interference and reverberation [1], [2]. Classical beamforming, such as delay-and-sum and minimum variance distortionless response (MVDR), is effective when spatial statistics and array geometry are well estimated; however, performance often degrades in dynamic acoustics or under array mismatch where covariance estimation becomes inaccurate.

Deep learning has advanced multichannel SE through two main strategies. The first integrates neural mask estimation with beamforming, where time-frequency masks guide spatial covariance estimation and beamformer weights. Erdogan *et al.* [3] estimate speech/noise spatial covariance matrices (SCMs) from masks and then compute MVDR-like filters, improving robustness over analytical pipelines yet leaving performance tightly coupled to SCM accuracy. The second line builds end-to-end models that map multichannel inputs to

enhanced speech via convolutional, recurrent, or transformer-style encoders. Pandey *et al.* [4] strengthen long-range temporal dependencies with a temporal-path attention recurrent network but fuse channels implicitly without per-time-frequency spatial beamforming. Li *et al.* [5] combine convolution and transformers to improve spectral-temporal modeling at the cost of wider attention windows or large receptive fields. Conformer-based encoders [6] and transformer separators [7] capture global context via full self-attention, but the quadratic complexity and the lack of explicit microphone-space reasoning pose challenges for real-time multichannel deployment.

Graph neural networks (GNNs) model a microphone array as a graph, channels as nodes and inter-channel relations as edges, so that spatial dependencies are learned from data [8]. While complex-valued and temporal extensions further enrich modeling [9], graph structures are frequently fixed or derived from static correlations and thus adapt poorly to changes in acoustics or array geometry. Recent works integrate graph attention or channel-selection mechanisms for ad-hoc arrays to better handle variable topology and channel count [10]. GNNs have also been leveraged for distributed-array localization, underscoring the value of graph-conditioned spatial reasoning under topology variability [11]. Meanwhile, dense frequency-time attentive networks combining spectral transformers with temporal conformers [12], [13] remain competitive, but they incur high computational cost and latency while still decoupling spatial and temporal cues.

Motivated by these limitations, we propose an end-to-end multichannel SE framework that performs per-time-frequency spatial filtering on a microphone graph, followed by efficient temporal modeling with localized self-attention and complex-mask refinement. Specifically, the spatial front-end outputs real-valued convex channel weights (nonnegative and summing to one) to form a stable spatially filtered reference stream, and we relate these weights to the relative magnitude profile of the MVDR numerator under narrow-band assumptions, while leaving complex sign/phase correction to the mask head. The graph formulation parameterizes spatial processing through inter-channel relations rather than explicit microphone coordinates; in the experiments below, it is instantiated on a four-microphone circular array using the corresponding graph

connectivity. On the spatialized WSJCAM0 and DNS-2020 datasets, the proposed method achieves consistent gains over strong baselines with lower computational cost.

II. PROPOSED METHOD

A. Overall Framework

Figure 1 summarizes the proposed end-to-end pipeline. Multichannel audio is mapped to the STFT domain and processed by a graph-attention spatial front-end to obtain a spatially filtered reference stream. A local-attention Conformer then captures short-/mid-range temporal structure with bounded look-ahead, and a compact U-Net predicts a complex ratio mask for residual magnitude/phase correction before waveform reconstruction. The system is trained end-to-end with the phase-constrained magnitude loss.

B. Spatial Feature Extraction Module

We operate in the multichannel STFT domain. Let $X_p(f, t) \in \mathbb{C}$ be the STFT representation of the signal from microphone $p \in \{1, \dots, P\}$ at frequency f and time frame t . For each (f, t) we build a directed, fully connected microphone graph whose nodes are channels. At layer ℓ , node i carries a real feature vector $\mathbf{h}_i^{(\ell)}(f, t) \in \mathbb{R}^D$ obtained by embedding $[\Re X_i(f, t), \Im X_i(f, t)]$. When clear, we omit (f, t) and write $\mathcal{N}(i)$ for the neighbor set (including self).

Our spatial module is a dual-transform graph-attention beamformer based on GATv2 [14]. Unlike the single linear map in GATv2, we use two projections to form receiver-specific queries and sender-specific keys:

$$\mathbf{q}_i^s = W_1^{(\ell)} \mathbf{h}_i^{(\ell)}, \quad \mathbf{k}_j^s = W_2^{(\ell)} \mathbf{h}_j^{(\ell)}, \quad (1)$$

with $W_1^{(\ell)}, W_2^{(\ell)} \in \mathbb{R}^{d \times D}$ and $\mathbf{q}_i^s, \mathbf{k}_j^s \in \mathbb{R}^d$. This query/key separation yields an asymmetric, reference-conditioned edge scoring, allowing the reference node to emphasize channels that are most informative at each time-frequency bin. A feed-forward scorer $F : \mathbb{R}^{2d} \rightarrow \mathbb{R}$ with LeakyReLU produces attention scores

$$e_{ij}^{(\ell)} = F([\mathbf{q}_i^s, \mathbf{k}_j^s]), \quad (2)$$

which are normalized over $\mathcal{N}(i)$:

$$a_{ij}^{(\ell)} = \frac{\exp(e_{ij}^{(\ell)})}{\sum_{k \in \mathcal{N}(i)} \exp(e_{ik}^{(\ell)})}, \quad a_{ij}^{(\ell)} \in [0, 1], \quad \sum_j a_{ij}^{(\ell)} = 1. \quad (3)$$

The softmax enforces a convex mixing constraint across channels, which stabilizes the spatial stage (gain-controlled) and avoids uncontrolled complex rotations before mask-based correction. We index spatial layers as $\ell = 0, \dots, L_s - 1$. We use the attention weights in two complementary ways. First, after the last spatial layer (indexed by $L_s - 1$), a per-frequency spatially filtered complex spectrum at a chosen reference output r is formed as

$$Y_r(f, t) = \sum_{j=1}^P a_{rj}^{(L_s-1)}(f, t) X_j(f, t), \quad (4)$$

which applies real, nonnegative, sum-to-one weights across channels; complex phase/sign is deferred to the mask head. Second, for representation learning we update node features via message passing:

$$\mathbf{h}_i^{(\ell+1)}(f, t) = \sigma\left(\sum_{j \in \mathcal{N}(i)} a_{ij}^{(\ell)}(f, t) W_v^{(\ell)} \mathbf{h}_j^{(\ell)}(f, t)\right), \quad (5)$$

where $W_v^{(\ell)} \in \mathbb{R}^{D \times D}$ and $\sigma(\cdot)$ is a pointwise nonlinearity. The pair $(Y_r, \mathbf{h}_r^{(L_s)})$ is forwarded to the temporal encoder.

This dual-transform design conditions the score on both the receiver i and the sender j , enabling finer inter-channel modeling without explicit covariance estimation. Under narrow-band processing, we interpret the learned convex weights for the reference output in a restricted MVDR-consistent sense. Let $\mathbf{w}_{\text{MVDR}}(f) \in \mathbb{C}^P$ denote the conventional MVDR weight vector at frequency f [15], given by

$$\mathbf{w}_{\text{MVDR}}(f) = \frac{\Phi_{nn}^{-1}(f) \mathbf{d}(f)}{\mathbf{d}^H(f) \Phi_{nn}^{-1}(f) \mathbf{d}(f)}, \quad (6)$$

where $\Phi_{nn}(f)$ is the noise covariance matrix, $\mathbf{d}(f)$ is the desired-source steering vector, and $(\cdot)^H$ denotes the Hermitian transpose. Since $a_{rj}(f, t)$ is real, nonnegative, and normalized, it is not intended to reproduce the full complex MVDR filter; instead, it provides an MVDR-inspired normalized magnitude profile of the MVDR numerator,

$$a_{rj}(f, t) \approx \frac{|[\Phi_{nn}^{-1}(f) \mathbf{d}(f)]_j|}{\sum_{k=1}^P |[\Phi_{nn}^{-1}(f) \mathbf{d}(f)]_k|}. \quad (7)$$

The downstream cRM head then compensates for residual phase/sign and reverberation-related errors.

C. Temporal Sequence Modeling Module

The temporal encoder adopts a Conformer-style block [6] and operates on the spatial outputs of Sec. II-B. Let $Y_r(f, t) \in \mathbb{C}$ be the spatially filtered spectrum at the reference output r , and let $\mathbf{h}_r(f, t) \in \mathbb{R}^D$ be its spatial embedding forwarded from the last spatial layer (we henceforth write $\mathbf{h}_r(f, t) \triangleq \mathbf{h}_r^{(L_s)}(f, t)$ for brevity).

For each frequency bin f , we form a time sequence

$$\mathbf{x}_t(f) = [\Re Y_r(f, t), \Im Y_r(f, t), \mathbf{h}_r(f, t)] \in \mathbb{R}^{D_{\text{in}}},$$

$$D_{\text{in}} = 2 + D,$$

where $t \in \{1, \dots, R\}$ indexes frames and R is the number of frames in the current chunk (i.e., a block of consecutive frames). All temporal parameters are shared across frequency bins, while sequences are processed independently for each f .

To bound latency and computation, self-attention is restricted to a symmetric local window of $\pm w$ frames (single-head notation for clarity). Let d be the projection dimension and define $\mathbf{q}_t = W_q^{(T)} \mathbf{x}_t(f)$, $\mathbf{k}_\tau = W_k^{(T)} \mathbf{x}_\tau(f)$, $\mathbf{v}_\tau = W_v^{(T)} \mathbf{x}_\tau(f)$ with $W_q^{(T)}, W_k^{(T)}, W_v^{(T)} \in \mathbb{R}^{d \times D_{\text{in}}}$. With the window set $\mathcal{W}(t) = \{\tau : \max(1, t - w) \leq \tau \leq \min(R, t + w)\}$, the localized attention output is

$$\mathbf{y}_t = \sum_{\tau \in \mathcal{W}(t)} \frac{\exp(\mathbf{q}_t^\top \mathbf{k}_\tau / \sqrt{d})}{\sum_{\kappa \in \mathcal{W}(t)} \exp(\mathbf{q}_t^\top \mathbf{k}_\kappa / \sqrt{d})} \mathbf{v}_\tau, \quad (8)$$

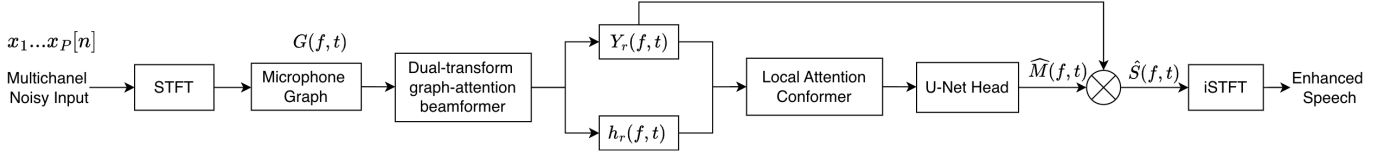


Fig. 1. Overall pipeline of the proposed MVDR-consistent graph-attention beamformer with a local Conformer and complex-mask refinement.

where $\mathbf{y}_t \in \mathbb{R}^d$. This reduces the per-layer attention cost from $O(R^2d)$ to $O(Rwd)$ and induces a fixed look-ahead of w frames. Moreover, stacking several localized layers helps recover mid-range context.

The Conformer convolutional module is implemented with a depthwise-separable bottleneck to further reduce parameters and multiply-accumulate operations per second (MAC/s) [16]. With bottleneck dimension $d_r < d$, kernel size k , nonlinearity $\phi(\cdot)$, and projections $W_1 \in \mathbb{R}^{d_r \times d}$, $W_2 \in \mathbb{R}^{d \times d_r}$, we use

$$\mathbf{z}_t = W_2 \left(\text{DepthConv}_k(\phi(W_1 \mathbf{y}_t)) \right), \quad (9)$$

where DepthConv_k denotes channel-wise 1-D convolution along time with kernel size k .

To further compress large linear maps, we apply low-rank factorization

$$W \approx UV, \quad U \in \mathbb{R}^{d \times r_{\text{LR}}}, \quad V \in \mathbb{R}^{r_{\text{LR}} \times d}, \quad r_{\text{LR}} < d, \quad (10)$$

which preserves expressiveness while reducing the number of parameters and processing cost. The resulting temporal features $\mathbf{z}_t$ (for each f) are then fed to the mask head for cRM prediction and subsequent iSTFT reconstruction.

D. Mask Prediction and Speech Reconstruction

The mask head refines the spatially filtered stream produced by the spatial and temporal modules. Given the reference spectrum $Y_r(f, t) \in \mathbb{C}$ and its temporal features, a lightweight U-Net predicts a complex mask $\hat{M}(f, t) \in \mathbb{C}$ (aligned with the complex ratio mask concept [17]). The enhanced spectrum is then obtained by element-wise multiplication:

$$\hat{S}(f, t) = \hat{M}(f, t) Y_r(f, t).$$

The time-domain signal $\hat{s}[n]$ is reconstructed via inverse short-time Fourier transform (iSTFT) with overlap-add.

This design explicitly decouples coarse spatial weighting (real, nonnegative channel weights) from fine complex correction in the mask head, preserving phase fidelity without explicit covariance estimation. We train the whole system end-to-end with the phase-constrained magnitude (PCM) loss [18], computed between the iSTFT-reconstructed waveform $\hat{s}[n]$ and the clean reference $s[n]$. Making the reconstruction path differentiable allows the mask to compensate for the missing complex sign/phase introduced by the spatial stage.

III. EXPERIMENTAL SETUP

A. Datasets

We evaluate on two spatialized multichannel datasets generated by convolving clean speech and noise with simu-

lated room impulse responses (image-source method). A four-microphone circular array (radius 10 cm) is used throughout; all signals are resampled to 16 kHz. Speech and noise are convolved with independent impulse responses before mixing. To avoid leakage, room geometries, impulse-response sets, and source-array configurations are disjoint across training/validation/test; sources are at least 0.5 m from walls, source-array distance is sampled in [0.75, 2] m, and reverberation time is sampled in [0.2, 1.2] s.

For WSJCAM0, clean speech is drawn from WSJ-CAM0 [19] and background noises follow the REVERB challenge protocol [20]; mixtures are generated with SNR in [5, 20] dB. For DNS-2020 [21], which is originally single-channel, we use the same array geometry and room simulation procedure to generate spatialized mixtures; speech/noise segments are 3–6 s for training and 3–10 s for validation/testing, with SNR in [−10, 10] dB, so the convolved background sources act as direction-dependent non-speech interferers.

B. Experimental Settings

Inputs are transformed to the complex STFT domain using a 32 ms rectangular window (512 samples) and 75% overlap (8 ms hop, 128 samples), yielding 257 frequency bins per frame. We adopt chunked streaming with chunk length $R = 126$ frames and shift $S = 63$ frames, followed by overlap-and-stitch on the waveform.

The dynamic microphone graph is fully connected with self-loops. Attention weights from the last spatial layer form the per-frequency reference stream $Y_r(f, t)$ (Sec. II-B), and the corresponding spatial embedding is fed to the temporal encoder (Sec. II-C). Local self-attention uses a fixed window of $\pm w$ frames with $w = 4$ (9 frames ≈ 72 ms context and ≈ 32 ms look-ahead at an 8 ms hop); together with the depthwise temporal convolution receptive field, this yields bounded algorithmic latency in streaming inference.

Model width is set to a feature dimension $D = 128$ across modules. Training uses randomly cropped 4 s segments (batch size 8) with Adam (initial learning rate 4×10^{-4} , halved after 5 epochs) and the PCM loss on the iSTFT-reconstructed waveform. Unless otherwise specified, we use $r = 1$ as the reference stream, a single attention head, $d = D$ with low-rank bottleneck $r_{\text{LR}} = D/2$, $L_s = 2$ spatial attention layers, $L_t = 6$ local-Conformer blocks, and an odd depthwise temporal kernel size $k = 15$. We report SI-SDR (dB), PESQ, and STOI (%), as well as the number of parameters (M) and computational complexity (MAC/s).

TABLE I
EXPERIMENTAL RESULTS ON THE SPATIALIZED WSJCAM0 AND DNS-2020 DATASETS.

Method	WSJCAM0			DNS-2020			Complexity	
	SI-SDR	PESQ	STOI	SI-SDR	PESQ	STOI	Param (M)	MAC/s (G)
Unprocessed	-3.8	1.38	70.9	-7.6	1.38	63.8	—	—
FasNet-TAC [22]	8.2	2.93	94.7	4.7	2.26	86.5	2.9	55.3
ADCN [23]	12.0	3.42	97.3	7.8	2.84	92.3	9.3	72.8
TPARN [4]	10.4	3.43	96.9	8.4	2.75	91.9	3.2	71.8
ADCN-TPARN [23]	13.8	3.64	98.0	10.0	2.99	93.7	40.0	360.0
DeFT-AN [12]	15.7	3.63	98.1	9.9	3.01	92.4	2.7	95.6
Ours	16.3	3.69	98.2	10.3	3.05	92.7	3.1	45.4

IV. EXPERIMENTAL RESULTS

We evaluate our method on both the spatialized WSJCAM0 and DNS-2020 datasets. In the following, we present a performance comparison with state-of-the-art baselines and conduct an in-depth ablation study.

A. Overall Performance Comparison

Table I compares our model with recent multichannel baselines on the spatialized WSJCAM0 and DNS-2020 datasets. On WSJCAM0, our system achieves a SI-SDR gain of +5.9 dB over TPARN and consistent perceptual gains over DeFT-AN, demonstrating that dynamic per-time-frequency spatial weighting with localized temporal encoding effectively reduces distortion while improving quality and intelligibility. On the more challenging DNS-2020 benchmark, we observe consistent improvements across all metrics over both TPARN and DeFT-AN, confirming robustness under varying SNRs and room conditions.

The model remains lightweight and computationally efficient. Its measured complexity is 45.4 G MAC/s, about half that of DeFT-AN. This matches the expected $O(Rwd)$ scaling enabled by localized attention and low-rank, depthwise-separable convolutions in the temporal block.

B. Ablation Studies

Table II presents an ablation study on the WSJCAM0 dataset, evaluating the contribution of each component and the effect of various architectural configurations.

(A) *Removal of Sub-Modules*: In sub-S, the first microphone STFT replaces the spatial module; sub-T bypasses the local-Conformer encoder; and sub-M removes the cRM head. The results show that temporal modeling contributes the largest gain, followed by spatial fusion and mask refinement, consistent with their roles in suppressing smearing, structured interference, and residual complex errors.

(B) *Module Order*: Our full model adopts the S-T-M order (spatial filtering → temporal encoding → complex-mask refinement). Among the two permutations (S-M-T and T-S-M), placing the mask before the temporal encoder performs better, as early spatial fusion provides cleaner cues for downstream modules. Nevertheless, S-T-M achieves the best overall results.

(C) *Spatial Network Variations*: Among various spatial backbones (GCN/GAT/GIN), the standard GATv2 performs

TABLE II
ABLATION ON SPATIALIZED WSJCAM0. “SUB-S/T/M” DENOTE REMOVAL OF THE SPATIAL/TEMPORAL/MASK MODULE, RESPECTIVELY. THE DEFAULT PIPELINE IS S-T-M; **PROPOSED** DENOTES THE FULL MODEL (DUAL-TRANSFORM GATv2 SPATIAL BACKBONE + LOCAL(± 4) CONFORMER).

Variations	SI-SDR	PESQ	STOI	Param (M)	MAC/s (G)
(A) Remove Sub-Module					
sub-S	12.1	3.40	95.9	2.2	31.3
sub-T	6.1	3.15	92.6	2.3	29.9
sub-M	14.7	3.57	97.6	3.1	44.6
(B) Module Order					
S-M-T	14.3	3.53	97.2	3.1	44.8
T-S-M	13.9	3.49	96.8	3.1	44.8
(C) Spatial Backbone					
GCN	12.6	3.46	96.3	2.8	39.1
GAT	13.4	3.51	96.9	2.9	40.1
GIN	13.8	3.52	97.0	2.9	40.5
GATv2 (standard)	14.5	3.58	97.5	3.0	44.3
(D) Conformer (Local Window)					
Global (MHA)	14.2	3.55	97.3	3.1	62.7
Local (± 2)	14.9	3.59	97.6	3.1	44.8
Local (± 6)	15.8	3.64	98.1	3.1	46.0
Proposed	16.3	3.69	98.2	3.1	45.4

the best, indicating that asymmetric, data-driven inter-channel relations are key. Our dual-transform GATv2 further improves performance when coupled with the temporal module and complex mask, highlighting the benefit of fine-grained attention for spatial filtering in dynamic acoustic environments.

(D) *Conformer Local Window*: Localized attention improves accuracy while reducing compute compared with global attention. The default ± 4 window offers a good accuracy-efficiency balance for streaming, while larger windows yield diminishing returns since the most useful context is short- to mid-range, with depthwise convolutions refining local structure.

V. CONCLUSION

We presented an end-to-end multichannel speech enhancement model that unifies a dual-transform graph-attention beamformer with a local-attention Conformer and a complex-mask head. The resulting real, convex spatial weights provide an MVDR-inspired normalized magnitude profile of the MVDR numerator under narrow-band processing. Across diverse simulated acoustic conditions, the method delivers consistent gains over strong baselines with substantially lower complexity. Ablation studies further verify that all components provide complementary gains in distortion reduction and perceptual quality.

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